

Open-source document OCR for clinical scans

A model bake-off, distillation study, and layout/bbox evaluation

328 synthetic clinical scans + 651 real medical scans · 2× RTX 4090 · all local, no PHI

The question

What should replace Tesseract / commercial layout-OCR for scanned clinical documents?

- Scanned referrals, lab panels, discharge summaries, faxes — degraded, rotated, handwritten, tabular.
- Need: (1) the values (MRNs, dates, lab decimals, meds) AND (2) the structure (tables, reading order, selection controls).
- Constraint: cheap enough to run at document volume — target < \$0.01 / page.
- Open-weight VLMs now exist that do full-page transcription in one pass. Do any of them actually work here?

The models

12 open-weight OCR / document VLMs, one incumbent, comparable sizes and licences

model	params	licence
Qwen3.8-27B-FP8	27B (FP8)	verify
Qwen2.5-VL-7B	7B	Apache-2.0
Qwen2.5-VL-3B	3B	Apache-2.0
dots.mocr	3B	MIT
dots.ocr	3B	MIT
olmOCR-2	8B	Apache-2.0
PaddleOCR-VL	0.9B	Apache-2.0
DeepSeek-OCR	3B MoE	MIT
Chandra OCR 2	5B	OpenRAIL-M
granite-docling	0.26B	Apache-2.0
Nanonets-OCR2	3B	Apache-2.0
Tesseract v5	-	Apache-2.0

Corpus 1 — ClinOCR-Bench (the baseline)

328 documents · 6 degradation subsets · human-audited ground truth



Patient: John Smith DOB: 05/20/1965 MRN: 789012
Accession #: A583920-25



**SPRINGFIELD
GENERAL
HOSPITAL**

Sunrise Diagnostics Phone:
North Valley Imaging Center 456
Oak Avenue, Suite 200 Anytown,
USA 12345
(555) 123-4567

Patient: John Smith
DOB: 05/20/1965
MRN: 789012
Gender: Male Referring
Physician: Dr. Emily White

Visit #: 456-789 Date of Visit: 06/18/2025

Diagnostic Imaging Report - Chest X-ray (2 views)

Procedure: Radiography of the chest, posteroanterior and lateral views.

Clinical Indication: Persistent cough and shortness of breath.

Findings: The lungs are clear without evidence of consolidation, infiltrate, or effusion. The cardiomedial silhouette is within normal limits. The osseous structures of the thorax are unremarkable.

Impression: No acute cardiopulmonary process.

Scheduled appointments:
Follow-up appointment with Dr. White scheduled for 06/25/2025 at 11:00 AM to discuss the results and further management.

Date of Dictation: 06/18/2025 Technician: Sarah Jenkins, RT(R)

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normal

- subsets: normal · handwriting · poor · rotated · tables · mixed
- (shown: clean + handwriting; the tables subset is actually filled forms)

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handwriting

Corpus 2 — medreal (real scans)

651 pages assembled from public medical-document datasets, labelled by a vision LLM

source	pages	what it is
noisy-med	400	patient account statements, full page
prescription	200	prescription scans
india-hist	46	real 1878 hospital reports (external GT)
medform	5	medical forms

- Labels minted with DeepSeek-v4.1-flash (vision) at \$0.0009/page — the same prompt used in training
- Caveat: new-set ground truth is the teacher label, so it measures agreement, not human-verified accuracy
- Cheap generalisation check: a different pipeline, real scanners, non-English names/addresses

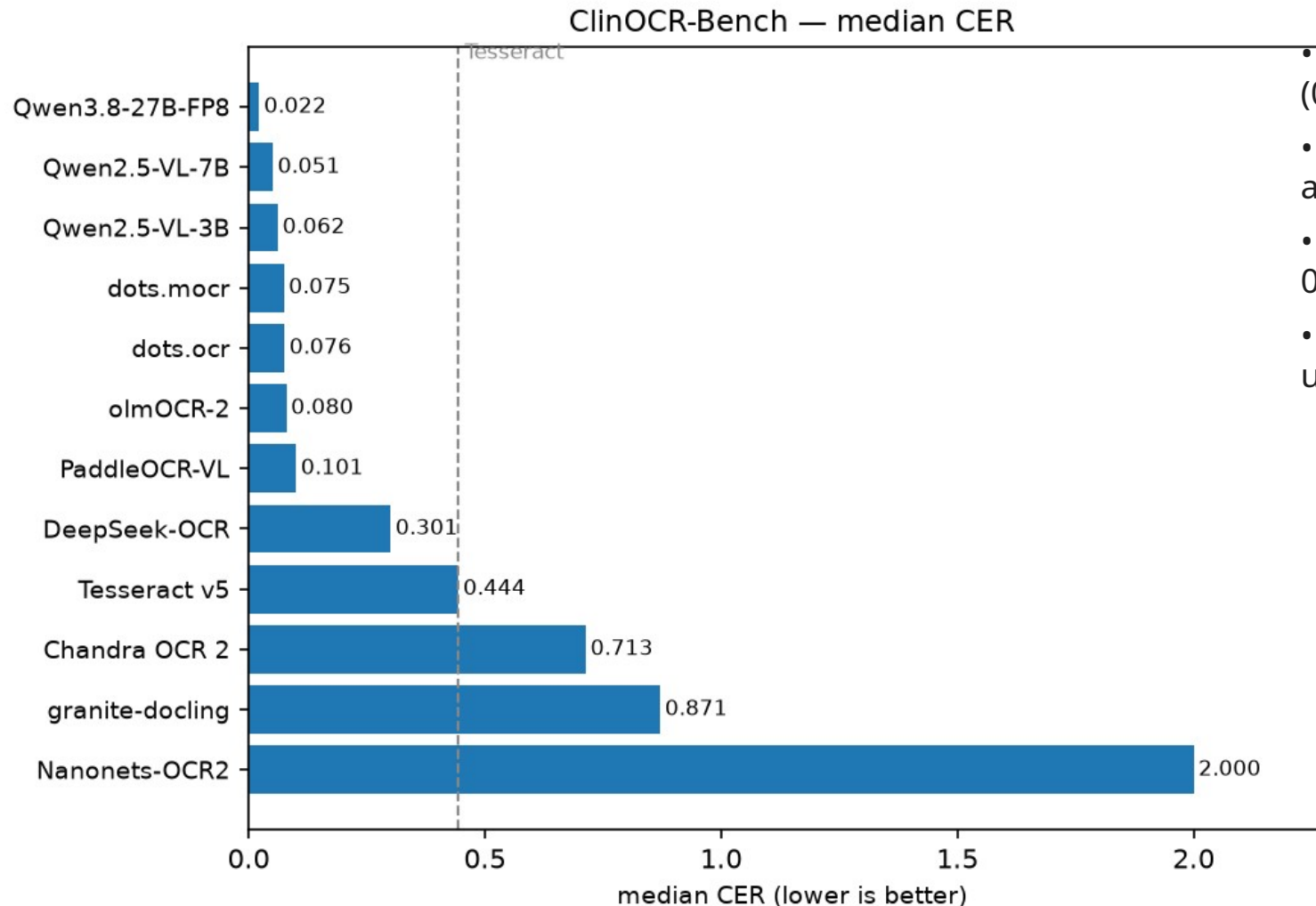
How we measure

Character error rate, with the pitfalls accounted for

- Primary metric: median CER (character error rate) vs ground truth — lower is better.
- Mean CER is NOT stable: a few unreadable pages loop into repeated text and pin at the CER cap of 2.0.
 - The runaway tail carries ~half the error and varies run to run — so we rank on median, and report runaways.
- Also measured: field-token recall/precision (dates, MRNs, decimals, codes), failure taxonomy, throughput, cost.
- Noise floor measured from n=5 identical repeats: median CER ± 0.003 , mean CER ± 0.023 .
 - Any difference smaller than that is not a finding.

Results — ClinOCR-Bench (328 docs)

median CER; Tesseract baseline dashed



- Every VLM beats Tesseract on degraded scans (0.02–0.10 vs 0.444).
- Qwen3.8-27B is a tier above everything (0.0224) and leads field recall (0.91).
- dots.mocr / Qwen7B / olmOCR-2 cluster at ~0.05–0.08 — cheap and strong.
- Chandra/granite/DeepSeek/Nanonets are not usable here.

Selection controls + bounding boxes

dots.mocr, native layout prompt — regions with bbox + category, form read as HTML tables

Date: 04/14/2026

Therapist Names: Mr. David Osei,
RBT / Ms. Julia Park, BCBA

Patient: Ethan C. Brennan

DAILY REPORT

MOOD AFFECT	☐ Anxious	☐ Calm / Content	☒ Frustrated	Came With	☐ Cup
	☐ Angry	☐ Cheerful / Happy	☐ Grumpy		☒ Backpack
	☐ Annoyed	☐ Energetic	☐ Tired		☐ Jacket
	☐ Loving				☒ Lunch Box
					☐ Other

AGGRESSION	STEMMING	NON COMPLIANCE	OTHER
☒ Hitting	☐ Thumb Sucking	☐ Non Compliance	☐ Inappropriate Laughing
☐ Kicking	☐ Mouthing	☒ Tantruming	☐ Genital Grabbing
☐ Hair Pulling	☐ Pica	☐ Elopting	☐ Inappropriate Touching
☐ Throwing	☒ Flopping	☐ Hand Flapping	☐ Verbal Aggression
☒ Pushing	☒ Screaming	☐ Hand Jerks	☒ Vocal Stereopathy
☐ Head Banging	☐ Crying	☒ Verbal Refusal	☐ Preservative Manding
☐ Scratching	☐ Repeating	☐ Climbing	☐ Side-Talk
☐ Pinching	☐ Licking	☐ Non-responsive	☐ Inappropriate Dance
☐ Grabbing	☐ Echolalia		
☐ Spitting	☐ Teeth Grinding		
☐ Biting	☐ Vomiting		
☐ Swiping			
☐ Self-Hitting			

LUNCH TIME	Eating Report:
	Ethan refused to eat initially. Accepted preferred crackers after a 10-minute motivating assessment. Ate approximately 50% of lunch with physical prompts.
	Targets Hit During Lunch:
	Manding for break (+2/5). Tact labels (0/5 — highly dysregulated). Following 1-step instructions (1/5).

GROUP SOCIAL	Today's Struggles and Strengths:
	High aggression in AM session related to schedule change. De-escalation via sensory break was effective. Strength: complied with hand-washing routine independently.

input (a filled behaviour form)

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dots.mocr regions (bbox + category)

Layout prompt: geometry + structure + controls

same model, dedicated prompt asking for JSON regions

run	bbox coverage	tables as HTML	text CER (median)	checked-box recall
dots.mocr (plain)	none	partial	0.0747	0.65
dots.mocr (layout)	100% (7091)	242 / 242	0.0274	0.70
dots.ocr (layout)	100% (6788)	242 / 242	0.0211	0.70

- → 100% of elements carry [x1,y1,x2,y2] + a category (Text, Section-header, Table, ...).
- → Every table returned as parsed HTML; text accuracy improves sharply vs the plain prompt.
- Caveat: ~0.70 of checked boxes captured, and the standalone-X 'Yes/No' forms are still missed (~0.0–0.10).
- Open item: does the geometry keep each check attached to the right field? Needs a structured-control scorer.

Do models capture selection controls?

recall of checked marks (☒/☒) on the form subset

model	checked-box recall	model	checked-box recall
Qwen2.5-VL-7B + 1 exemplar	0.85	dots.ocr	0.33
Qwen3.8-27B	0.82	Qwen2.5-VL-3B	0.24
olmOCR-2	0.77	PaddleOCR-VL	0.03
dots.mocr	0.65	Chandra / granite / DeepSeek / Tesseract	0.00

- Several open models reliably read ☒ vs ☐ as text — but as glyphs, not structured {control, state} fields.
- The models that emit coordinates (chandra, granite) capture none of the controls; the ones that capture controls mostly emit no coordinates.
- dots.mocr's layout prompt is the only run that gives both (see previous slide).

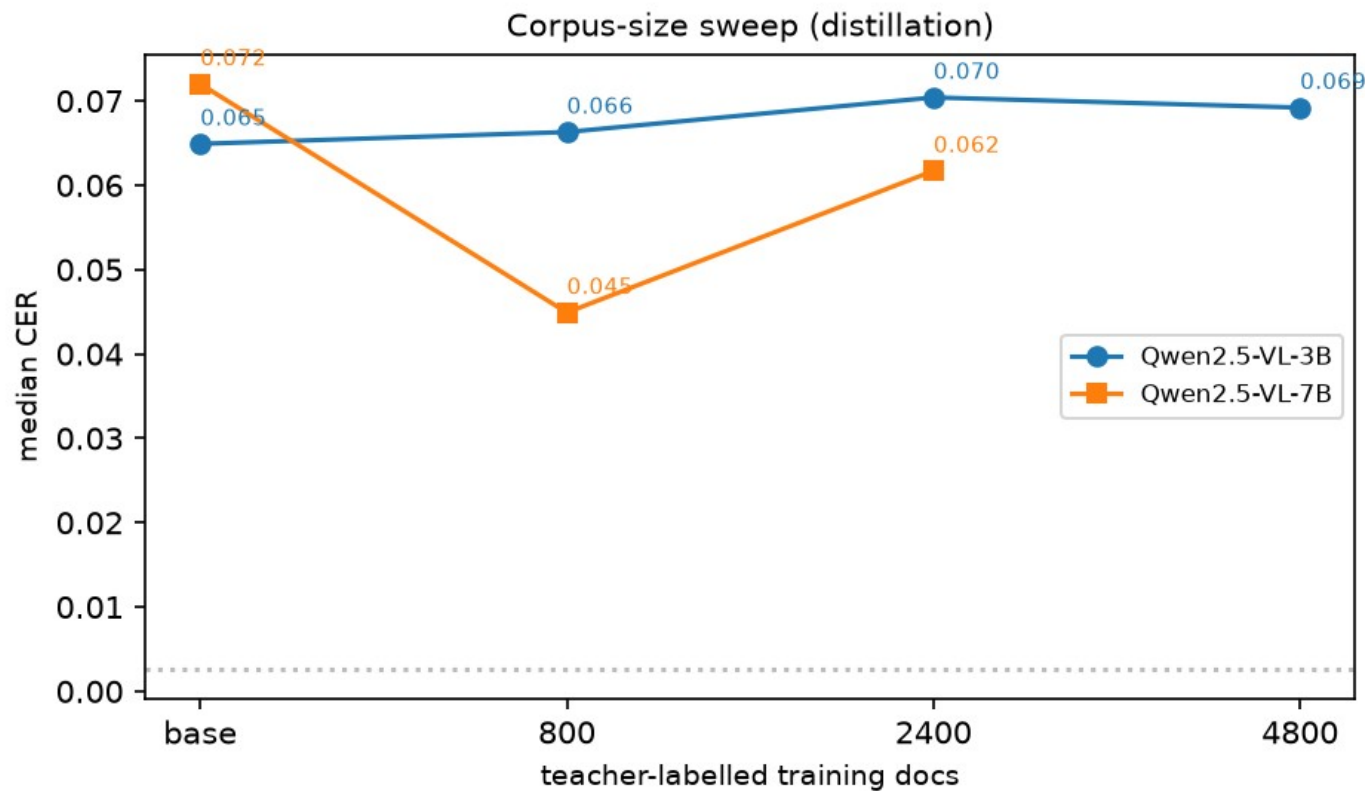
Training — distillation of a small student

LoRA on synthetic degraded clinical docs; teacher = Qwen3.8-27B

- Method: LoRA adapters ($r=16$, $\alpha=32$, dropout 0.05) on frozen Qwen2.5-VL base — NOT full fine-tuning.
- Targets q/k/v/o/gate/up/down (7 projections, incl. vision tower): 37M trainable on 3B (0.98%), 48M on 7B (0.57%).
- 1 epoch, lr $1e-4$ cosine, grad-accum 8, bf16, images capped at 802,816 px.
- Labels: either exact synthetic GT, or a teacher's transcript of unlabelled scans (no ground truth needed).
- 10 adapters trained: 4× (label-source study), 5× (corpus-size sweep), 1× (old+new mixed).

Training result — does more data help?

same teacher, same recipe, corpus size varied



- 3B: flat at/below the untuned base from 800→4800 docs → capacity limit, not data limit.
- 7B: 800 docs cuts median 0.072 → 0.045 (a real win); 2400 regresses to 0.062.
- Distillation helps the 7B, not the 3B; and more data is not monotonically better.
- Fine-tuning also raises runaway generation — a production caveat.

Cheaper alternative to fine-tuning: 1-shot exemplar

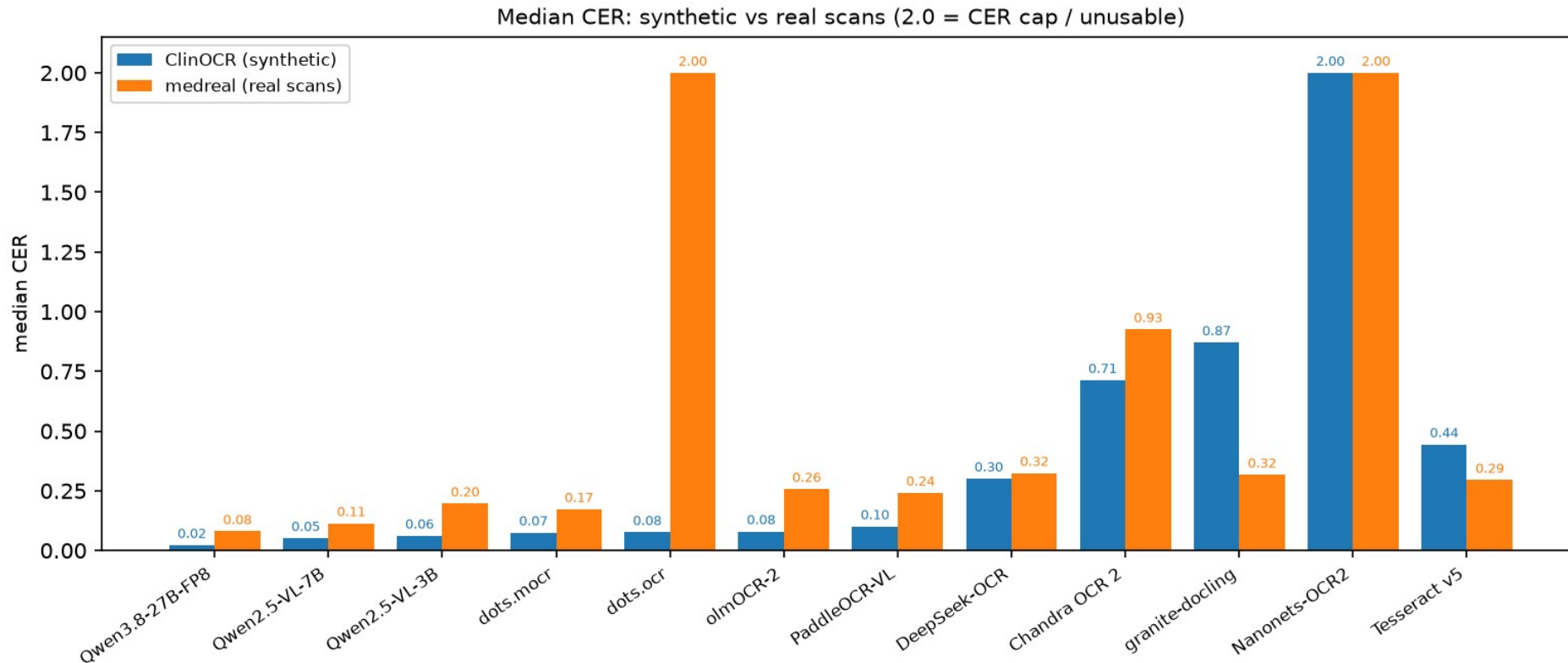
prepend one clean exemplar of the same form template

approach	median CER	cost / page
Qwen2.5-VL-7B (zero-shot)	0.0513	\$0.0021
Qwen2.5-VL-7B + 1 homogeneous exemplar	0.0206	\$0.0024
Qwen3.8-27B (teacher)	0.0224	\$0.0084

- One clean exemplar brings the 7B to ~27B accuracy on median CER — at ~1/4 the cost.
- No training required; the exemplar is a prompt-time addition.
- Trade-off: extra input tokens per request; distillation bakes the same benefit into the weights.

Generalisation — synthetic vs real scans

every model re-run on the 651-page real corpus



- Top of the ranking survives: Qwen3.8-27B (0.080) then Qwen2.5-VL-7B (0.113) on real scans too.
- Absolute error roughly doubles on real scans; real pages are harder but trigger fewer loops.
- dots.ocr collapses on real statements (median 2.0; 341/400 pages loop).
- granite-docling improves (0.87→0.32) but stays weak; PaddleOCR-VL degrades.

Cost — all viable models clear \$0.01/page

at \$4 / GPU-hour, you need only 0.111 pages/s per GPU

model (1 GPU unless noted)	pages/s	\$ / page	median CER
Qwen2.5-VL-3B	1.06	\$0.0010	0.0649
dots.mocr	0.67	\$0.0017	0.0747
Qwen2.5-VL-7B	0.54	\$0.0021	0.0513
Qwen2.5-VL-7B + 1-shot	0.46	\$0.0024	0.0206
Qwen3.8-27B (2 GPUs, TP=2)	0.26	\$0.0084	0.0224

- Cost is not the binding constraint — accuracy and structure are.
- Compute only; excludes retries/escalation. Real metric is \$ / acceptable page.
- 27B is under the line but with little headroom; int4 would roughly halve it.

What we'd recommend

for document intelligence: values + structure

- Primary: Qwen2.5-VL-7B (+1 exemplar) — near-27B accuracy at ~\$0.002/page, no training needed.
- Structure/geometry: dots.mocr with its layout prompt — bboxes + categories + HTML tables + most selection glyphs.
- Ceiling: Qwen3.8-27B — best values (field recall 0.91) and fewest runaways; still < \$0.01/page.
- Avoid as primary: PaddleOCR-VL, granite-docling, DeepSeek-OCR, Nanonets (values/runaways).
- Distillation: helps the 7B at ~800 docs; the 3B is capacity-limited. Fine-tuning is not a free win.

Caveats & next tests

what to verify before committing

- Selection-control association is unproven — we score glyph presence, not whether the check stays bound to the correct field.
- New-set ground truth is teacher-labelled (circular for the distilled model); only the 1878 scans have independent GT.
- $n=1$ per training point and a small form sample — repeat before quoting.
- Next: structured selection-control scorer; run the layout prompt on real filled forms; extend to real faxes at volume.